IPSec (IP Security, RFC 4301): Goals

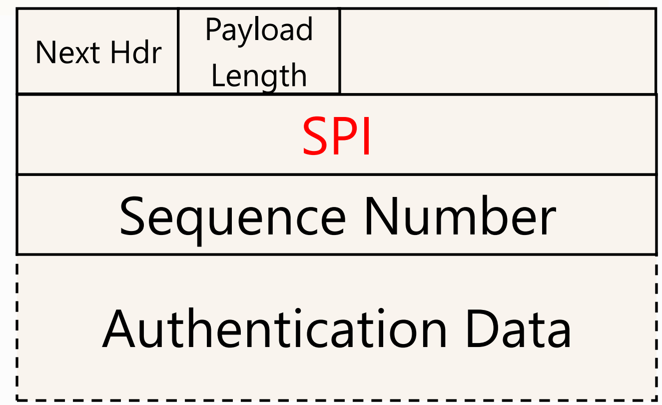
- Protect IP traffic
  - Datagram confidentiality
  - Datagram integrity control
- Two operational modes:
  - Transport mode
    - Uses / exposes original IP headers
    - Rarely used …
  - Tunnel mode
    - Encapsulates (possibly encrypting) original IP header

IPSec: Mechanisms

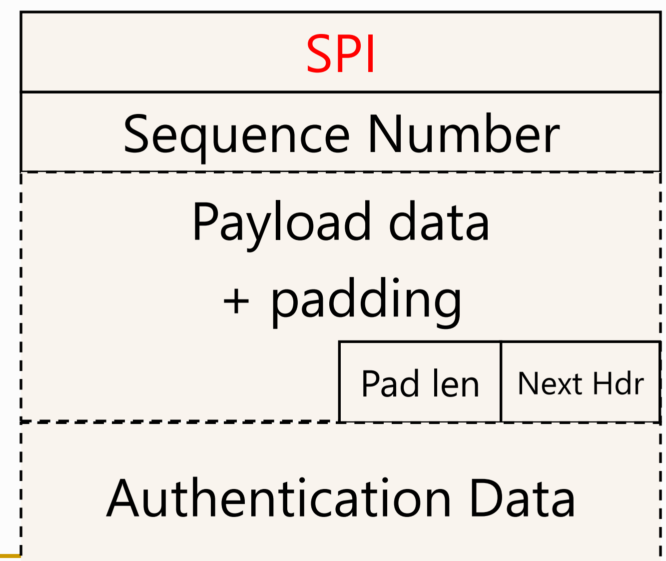
- Security Associations (SA, RFC 4301)
  - Security policies, mechanisms and crypto parameters used to secure the communication between a pair of hosts
  - Security Parameter Index (SPI)
    - SA identifier
    - Refers the SA that should be used to validate an IPSec datagram
- Extra optional fields for the IP header
  - Authentication Header (AH, RFC 4302)
    - Has an SPI
    - Keyed hash (MAC) of the whole IP datagram
  - Encapsulating Security Payload (ESP, RFC 4303)
    - Has an SPI
    - Authenticated cryptogram of the IP datagram payload

IPSec: AH and ESP mechanisms

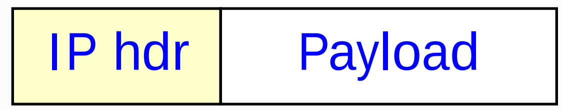
- ▷ AH goals
  - Connectionless integrity
  - Data origin authentication
  - Optional anti-replay service



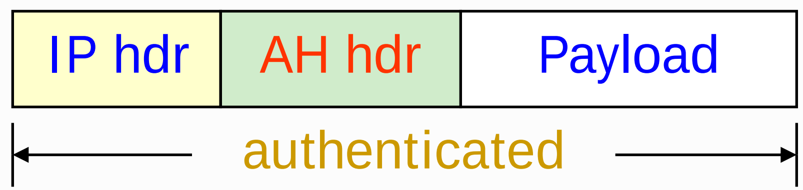
- ▷ ESP goals
  - Confidentiality (encryption)
  - Limited traffic flow confidentiality
  - Optional connectionless integrity
  - Optional data origin authentication
  - Optional anti-replay service



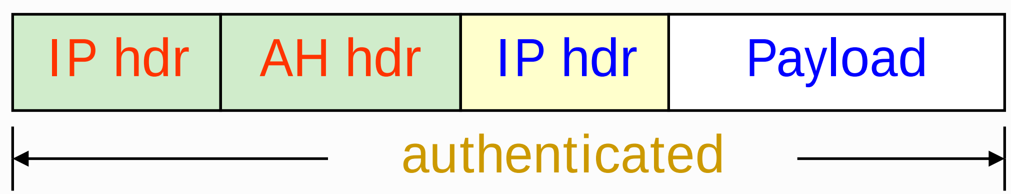
IPSec:
Authentication Header (AH)



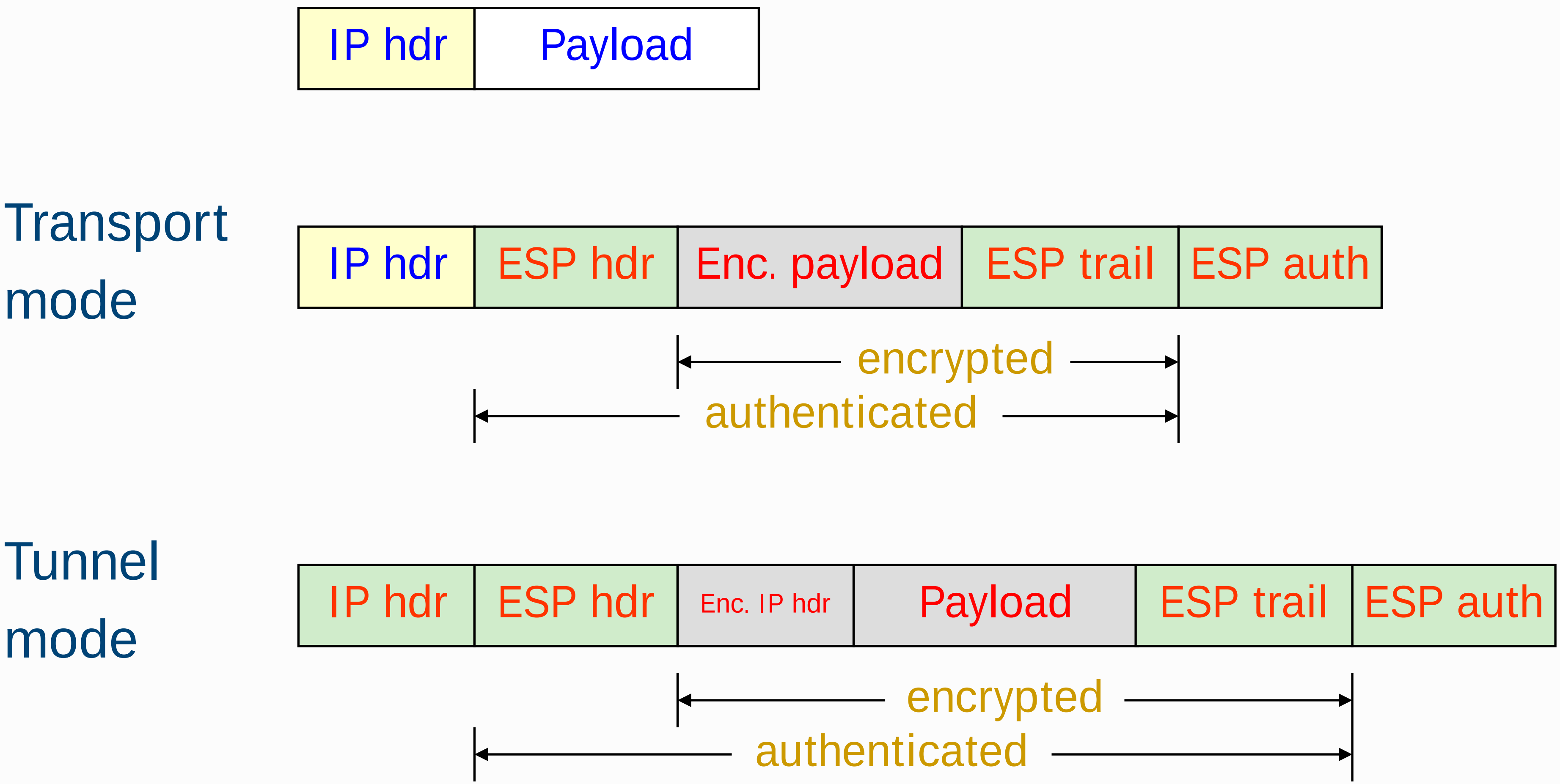
Transport mode



Tunnel mode



IPSec:
Encapsulating Security Payload (ESP)



IPSec: Operation

- If sender has an SA to destination IP:
  - Use AH and/or ESP according to SA
  - Changes the IP accordingly
    - Adds AH and/or ESP headers
    - Replaces plaintext header/payload by an encrypted version
- If receiver has an SA with the headers' SPI:
  - Validates IPSec headers according with their SA
  - Upon a validation failure the datagram is discarded
    - Silently

IPSec: SA and SPD databases

- SA database
  - Repository of local SAs
  - An SA is a bilateral peer agreement
    - A set of common rules to protect ID datagramas
    - But it only protects traffic in one direction!
- SPD database
  - Security Police Definition
  - A police states a protection level required
    - Traffic to peer P should be protected w/ mecanisms X, Y and Z

IPSec: Setup of SAs

- Manual
  - With line-oriented or graphical tools
  - With libraries
- Automatic with protocol
  - IKE (RFC 7296)

IPSec: Issues with NAT (RFC 3715)

- NAT interferes with the IP end-to-end paradigm
  - It is a “survival” hack
- NAT multiplexing impact is twofold
  - Network (changes src or dst IP)
  - Transport (changes src or dst port)
- Impact in IPSec
  - Doesn’t work with transport mode
    - AH prevents IP header changes,
    - ESP prevents transport header changes
  - Doesn’t work with tunnel mode
    - No transport ports to multiplex
- SPI-based is infeasible
- IKEv2 doesn’t work

Solution: NAT-T (NAT Traversal)

- In a nutshell, encapsulate IPSec packets in UDP packets
  - These can pass NAT boxes
  - In requires an UDP decapsulation prior to IPSec packet processing
- This is the basis of several other VPNs
  - They use some protocol to negotiate the SAs
  - Then use IPSec over UDP to protect packets
  - This is the case of OpenVPN & UA's Checkpoint VPN